

A disproof of conjecture 00000004289

Every almost-free group in the stated sense is flat-free

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1 Definitions and the assertion being refuted

Let G be an abelian group, regarded as a module over $\mathbb{Z}$, and let κ be any cardinal. The bilingual source defines the properties

$\text{AF}_\kappa(G) \iff G$ is not free and every subgroup $H \leq G$ with $|H| < \kappa$ is free,

$\text{FF}_\kappa(G) \iff$ every subgroup $H \leq G$ with $|H| < \kappa$ is flat.

Free means free as a $\mathbb{Z}$ -module, with an arbitrary basis and finite-support coordinates. Flat means that tensoring over $\mathbb{Z}$ preserves exact sequences, equivalently that it preserves injective module maps; see [1]. All subgroups are included, and the cardinal bound is strict in both definitions. The non-free requirement belongs to AF_κ , as in the source.

For clarity, keep the threshold and the group size separate:

$$E(\kappa, \lambda) \iff \exists G (|G| = \lambda \text{ and } \text{AF}_\kappa(G) \text{ and } \neg \text{FF}_\kappa(G)).$$

The usual reading of “at cardinal λ ” uses threshold $\kappa = \lambda$. On this reading the conjecture asserts that $\aleph_\omega$ is the least member of $S = \{\lambda : E(\lambda, \lambda)\}$. In particular, it asserts $E(\aleph_\omega, \aleph_\omega)$. We disprove this necessary existence clause and the minimum claim. The argument also covers a fixed threshold or a threshold unrelated to the group’s size.

2 Complete proof

Lemma 1. *Every free abelian group is a flat $\mathbb{Z}$ -module.*

Proof. If F has basis I , then $F \cong \bigoplus_{i \in I} \mathbb{Z}$. For every abelian group A there is a natural identification $A \otimes_{\mathbb{Z}} F \cong \bigoplus_{i \in I} A$. For an injective homomorphism $f : A \rightarrow B$, the induced tensor map becomes the coordinatewise map $\bigoplus_I f$. If its value on a finite-support tuple is zero, injectivity of f makes each coordinate zero. Thus the tensor map is injective. The injectivity criterion for flatness proves the lemma, including an empty or infinite basis. $\square$

Theorem 1. *For every cardinal κ and every abelian group G , $\text{AF}_\kappa(G)$ implies $\text{FF}_\kappa(G)$. Consequently $E(\kappa, \lambda)$ is false for all cardinals κ, λ .*

Proof. Assume $\text{AF}_\kappa(G)$. Let $H \leq G$ satisfy $|H| < \kappa$. The definition makes H free, so the lemma makes H flat. This proves $\text{FF}_\kappa(G)$ and contradicts the last condition in any proposed witness to $E(\kappa, \lambda)$. $\square$

3 Consequences and logical scope

There is no separating group at $\aleph_\omega$, or at any other cardinality. The set S is empty, so it has no least member. A least element must belong to the set; a mere lower bound or an infimum of an empty set would not express the source’s asserted minimum.

The separate $\aleph_1$ implication in the source is true, by the theorem with $\kappa = \aleph_1$, even without requiring $|G| = \aleph_1$. No regularity, cofinality, infinitude, or positivity assumption on the threshold is used. If the subgroup range is empty, the same implication still holds. The proof does not require the non-free conjunct, but the formal definition retains it exactly.

The source also asserts ZFC provability of the separation, a Shelah black-box construction, and independence over ZFC at every smaller cardinal. Refuting the necessary existence and minimum clauses disproves this compound assertion. This submission does not formalize a coded ZFC proof system, independence, or a black-box construction; a Lean theorem alone is not such a formalization. The unspecified range of “smaller cardinal” does not affect the universal nonexistence result.

4 Formal correspondence and verification

The project uses Lean 4.19.0 and Mathlib v4.19.0. The carrier has type `G : Type u`. Its `AddCommGroup G` structure provides the canonical integer action. Freeness is Mathlib’s basis-existence property, `Module.Free`. Flatness is its tensor-flatness class, `Module.Flat`, with proved injectivity and exactness interpretations over rings. The instance `Module.Flat.of_free` supplies the lemma. Subgroups are actual elements of `AddSubgroup G`, with inherited group operations and cardinality given by `Cardinal.mk`.

`AlmostFree`, `FlatFree`, `Separation`, and `SeparationExists` are exactly the predicates stated on page 1. `SeparationCardinals` is the set S . The theorem `no_separation_exists` quantifies independently over κ and λ in every universe. The final theorem `conjecture4289_disproof` states $\neg \text{IsLeast}(S, \aleph_\omega)$. The cardinal $\aleph_\omega$ is `Cardinal.aleph Ordinal.omega0`; $\aleph_1$ is `Cardinal.aleph 1`. Separate theorems exclude existence at $\aleph_\omega$, show S empty, and specialize the implication at $\aleph_1$.

The proof is symbolic and needs no external mathematical computation. From the pinned project in `lean/`, run

```
lake build
lake env lean -DwarningAsError=true Conjecture4289.lean
lake env lean -DwarningAsError=true Check.lean
```

The audit prints every authored definition, theorem type, and axiom dependency. Verification records cover all compiled proof-module constants, including generated ones, dependency identities, and the matching PDF. Local checks and independent internal review do not establish maintainer acceptance.

References

- [1] The Stacks Project, Section 10.39, *Flat modules and flat ring maps*, Definition 10.39.1 and Lemma 10.39.5. stacks.math.columbia.edu/tag/00H9.
- [2] Mathlib v4.19.0, `Mathlib/RingTheory/Flat/Basic.lean`, including `Module.Flat.of_free`. The exact dependency revision is recorded in the pinned project manifest.